



PROJECT DOCUMENTATION

Inventory Stock Alert System — StockSentinel

Subject: Java Programming

Project No.: 178

Program: B.Tech CSE (2025–29)

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Submission: *GitHub Repository*

1. Project Overview

StockSentinel is a Java Swing based Inventory Stock Alert System designed for a warehouse environment. The application stores product information, maintains current stock quantities, compares available stock with minimum stock levels, and identifies items that require restocking.

The system provides a simple graphical interface for registering products, entering stock information, searching for products, viewing low-stock alerts, and generating an overall stock report. The interface shown in the screenshots follows a dashboard-style layout with navigation on the left and the selected module displayed in the main workspace.

1.1 Problem Statement

A warehouse requires a system to monitor product stock and identify products that need restocking based on minimum stock levels. Manual monitoring can make it difficult to notice products that fall below their required threshold. The project addresses this requirement through automated comparison and visible stock alerts.

1.2 Objectives

- Store product names and quantities.
- Accept stock information and minimum stock levels.
- Compare available stock with minimum levels.
- Identify low-stock and out-of-stock products.
- Provide product searching functionality.
- Generate a consolidated stock report.
- Demonstrate Java arrays, Scanner input, loops, methods, searching, and conditional statements.

2. Technologies and Java Concepts

- Programming Language: Java
- User Interface: Java Swing
- Input Handling: Scanner-based inventory input as required by the case study
- Data Storage: Arrays for product names, quantities, and minimum stock levels
- Control Structures: for/while loops and if-else conditional statements
- Core Operations: registration, stock update, searching, comparison, alert generation, and reporting

2.1 Functional Modules

Module	Purpose	Main Function
Product Registration	Add new products and define minimum stock levels.	Product name + threshold
Stock Entry	Update current quantities and threshold values.	Update / delete product
Search	Find a specific product and view its stock status.	Name-based search
Low Stock Alerts	Display products below their minimum stock level.	Alert generation
Stock Report	Show all products with quantity, threshold, and status.	Summary report

3. Product Registration

The Product Registration module is used to add a new warehouse item. The user enters a product name and specifies the minimum stock level that should trigger stock attention. The interface also provides a button for console scanner input and a clear option.

The screenshot displays the 'Register Product' interface of the StockSentinel system. The sidebar on the left lists navigation options, with 'Register Product' currently selected. The main content area features a form with two input fields: 'PRODUCT NAME' containing 'Ps5' and 'MINIMUM STOCK LEVEL' containing '10'. Below the form are three buttons: a green '+ Add Product' button, a grey 'Use Console Scanner Input' button, and a grey 'Clear' button. The top of the interface shows the 'StockSentinel Warehouse Inventory Alert System' logo and a 'SYSTEM ACTIVE' status indicator. The bottom of the interface includes version information 'v1.0 - Java Swing' and the system name 'Java Swing - Inventory Alert System'.

Figure 3.1 — Product Registration screen showing PS5 with a minimum stock level of 10.

In the demonstrated screen, the product name is entered as “PS5” and the minimum stock level is set to 10. The Add Product button registers the item in the inventory.

3.1 Registration Flow

- Enter the product name.
- Enter the minimum stock level.
- Submit the product using Add Product.
- The product becomes available for stock management and reporting.

4. Stock Entry

The Stock Entry module allows an existing product to be selected and its quantity or minimum threshold to be updated. It also provides a delete option for removing an item from the inventory.

StockSentinel
Warehouse Inventory Alert System

SYSTEM ACTIVE

NAVIGATION

- Register Product
- Stock Entry**
- Search
- Low Stock Alerts
- Stock Report

Stock Entry
Update quantities, adjust threshold levels, or remove items

SELECT PRODUCT
Laptop

NEW QUANTITY
15

NEW MIN LEVEL
10

Update Stock Delete Product

v1.0 - Java Swing

Viewing: Stock Entry

Java Swing - Inventory Alert System

Figure 4.1 — Stock Entry screen showing Laptop with quantity 15 and minimum level 10.

The example shows Laptop selected with a new quantity of 15 and a minimum level of 10. The Update Stock button applies the entered values, while Delete Product removes the selected item.

4.1 Stock Update Logic

- Select an existing product from the product list.
- Enter the new quantity.
- Enter or modify the minimum stock level.
- Update the inventory record.
- Use the updated values for subsequent search, alert, and report operations.

5. Product Search

The Search module enables the user to find a specific product by name and view its current stock information. The result includes the product name, current quantity, minimum threshold, and the current stock status.

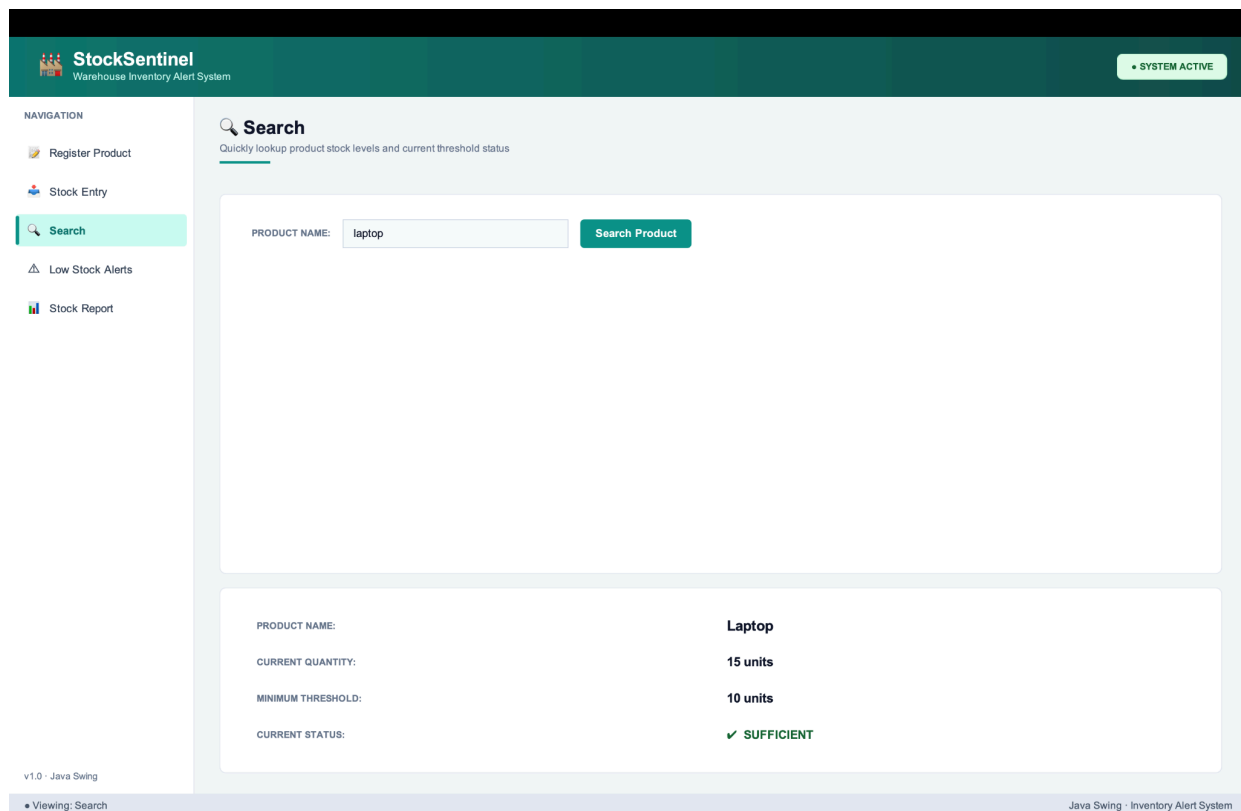


Figure 5.1 — Search screen showing Laptop with 15 units against a 10-unit minimum threshold.

In the example, the user searches for “laptop”. The system displays Laptop with 15 units in stock and a minimum threshold of 10 units. Since the current quantity is above the threshold, the displayed status is SUFFICIENT.

5.1 Search and Comparison

- Accept the product name from the user.
- Search the stored product names.
- Retrieve the corresponding quantity and minimum level.
- Compare quantity with the minimum threshold.
- Display the product's current status.

6. Low Stock Alerts

The Low Stock Alerts module identifies products that need immediate attention. Products below their minimum level are shown as LOW STOCK, while products with zero quantity are shown as OUT OF STOCK. The screen also displays the shortage amount.

The screenshot displays the StockSentinel Warehouse Inventory Alert System interface. The top header is dark green with the StockSentinel logo and 'Warehouse Inventory Alert System' text. A 'SYSTEM ACTIVE' button is in the top right. A left sidebar contains navigation links: 'Register Product', 'Stock Entry', 'Search', 'Low Stock Alerts' (highlighted), and 'Stock Report'. The main content area features a 'Check Alerts' button and a message: 'Found 3 item(s) requiring immediate attention!'. Below this is a table with four columns: 'Product Name', 'Current Qty', 'Status', and 'Deficit'. The table lists three items: Mouse (5 units, LOW STOCK, 15 units short), Keyboard (0 units, OUT OF STOCK, 15 units short), and Printer (0 units, OUT OF STOCK, 3 units short). The bottom status bar shows 'v1.0 · Java Swing' on the left and 'Java Swing · Inventory Alert System' on the right.

Product Name	Current Qty	Status	Deficit
Mouse	5	▲ LOW STOCK	15 units short
Keyboard	0	● OUT OF STOCK	15 units short
Printer	0	● OUT OF STOCK	3 units short

Figure 6.1 — Low Stock Alerts screen identifying three products requiring attention.

The demonstrated alert list contains Mouse with 5 units against a higher minimum level, Keyboard with 0 units, and Printer with 0 units. The interface reports three items requiring immediate attention and shows the corresponding shortage for each item.

6.1 Alert Conditions

- If current quantity is zero, the product is displayed as OUT OF STOCK.
- If current quantity is below the minimum threshold, the product is displayed as LOW STOCK.
- If current quantity meets or exceeds the minimum threshold, the product is considered SUFFICIENT and does not appear in the alert list.
- The shortage/deficit is calculated relative to the minimum required quantity.

7. Stock Report

The Stock Report provides a consolidated view of the warehouse inventory. It summarizes the total number of products and separates low-stock and out-of-stock items. A detailed table lists each product, its current quantity, minimum level, and status.

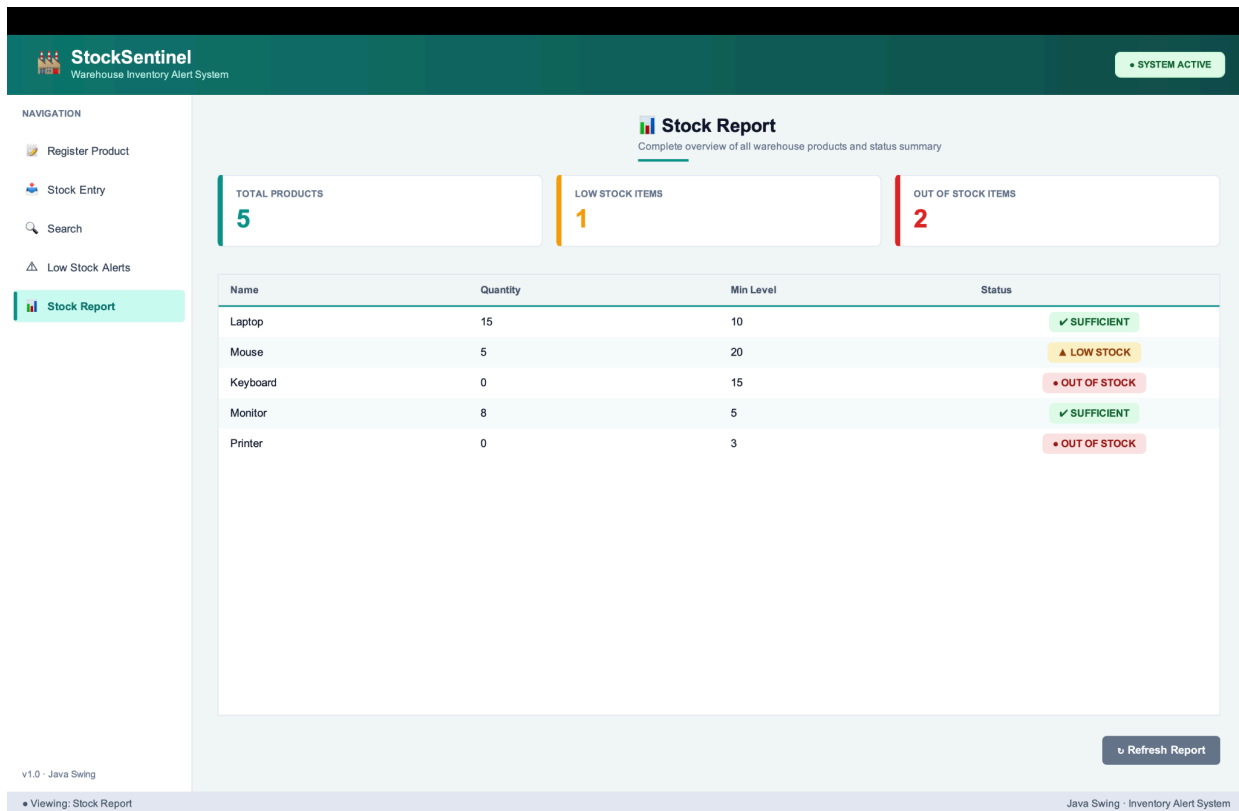


Figure 7.1 — Stock Report showing five products and their current inventory status.

The demonstrated report contains five products: Laptop, Mouse, Keyboard, Monitor, and Printer. The summary indicates 5 total products, 1 low-stock item, and 2 out-of-stock items. The table provides the quantity and minimum level used to determine each status.

7.1 Sample Report Interpretation

Product	Quantity	Minimum Level	Status
Laptop	15	10	SUFFICIENT
Mouse	5	20	LOW STOCK
Keyboard	0	15	OUT OF STOCK
Monitor	8	5	SUFFICIENT
Printer	0	3	OUT OF STOCK

8. Java Implementation Concepts

The case study specifically requires an implementation that demonstrates fundamental Java programming concepts. The completed application presents these concepts through inventory registration, stock processing, searching, comparison, and report generation.

- Arrays: used to organize product names, current quantities, and minimum stock levels.
- Scanner: used to accept inventory information as specified in the case study.
- Loops: used to process and examine multiple products.

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- If-else statements: used to compare available stock with minimum levels and determine status.
- Methods: used to separate operations such as stock analysis, searching, registration, and reporting.
- Searching: used to locate a specific product by its name.
- Conditional reporting: used to classify products as sufficient, low stock, or out of stock.

8.1 Overall Processing Flow

- Register products and their minimum stock thresholds.
- Enter or update current stock quantities.
- Process stored inventory records.
- Compare current quantities with minimum levels.
- Classify inventory status.
- Display search results, alerts, and the consolidated stock report.

9. Conclusion

The StockSentinel Inventory Stock Alert System provides a practical solution for monitoring warehouse inventory and identifying products that require restocking. The application combines product registration, stock updates, product searching, low-stock detection, and reporting in a single Java Swing interface.

The project demonstrates the required Java programming concepts, including arrays, Scanner input, loops, methods, searching, and conditional statements. The final interface also makes inventory status easy to understand by separating sufficient, low-stock, and out-of-stock products.

Overall, the completed system fulfills the objectives of Case Study 178 by providing a clear workflow for maintaining stock information and generating actionable inventory alerts.

10. GitHub Repository Link

The complete source code and project files for the StockSentinel Inventory Stock Alert System are available at the following GitHub repository:

Repository: <https://github.com/Yuvrajmishra-cell/Java-Project.git>

The repository contains the Java project source code and related project materials used to implement and demonstrate the inventory alert system.

Repository: Java-Project